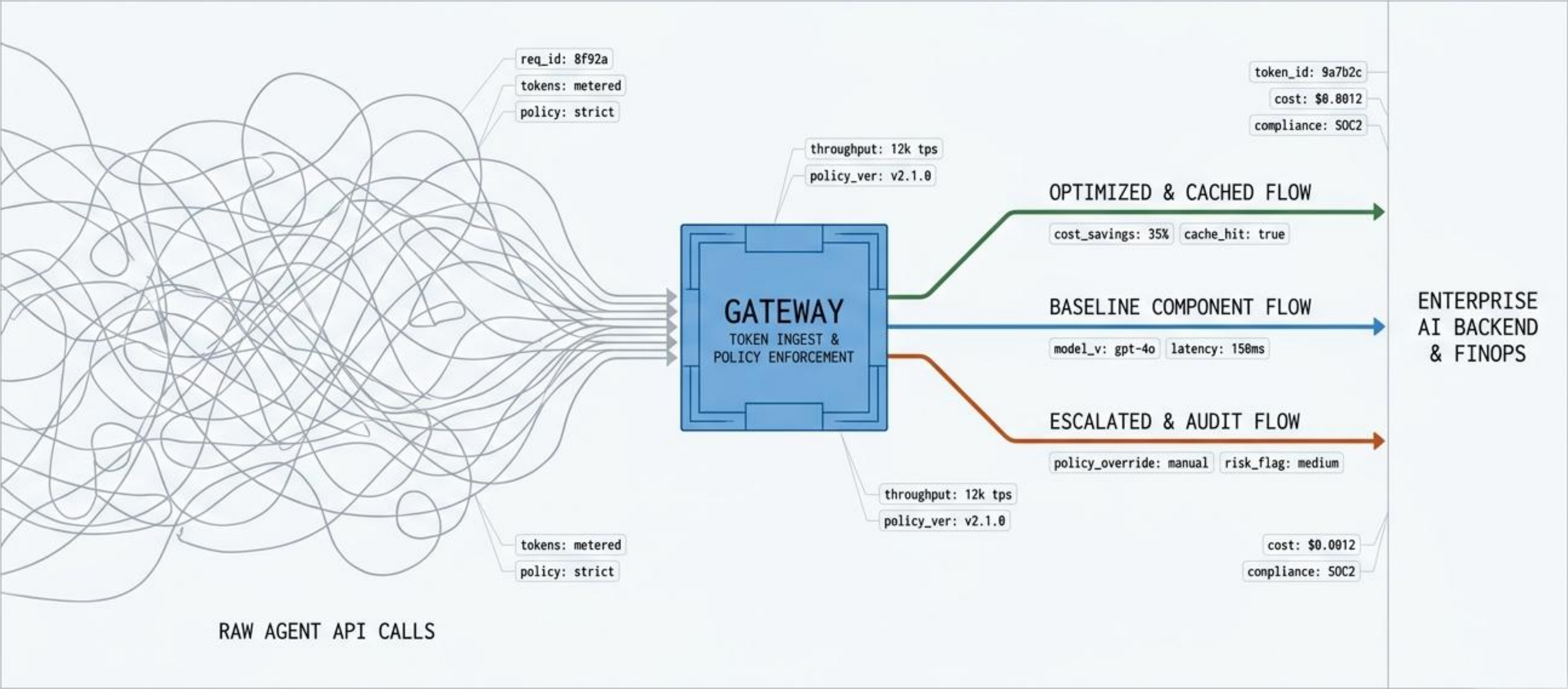
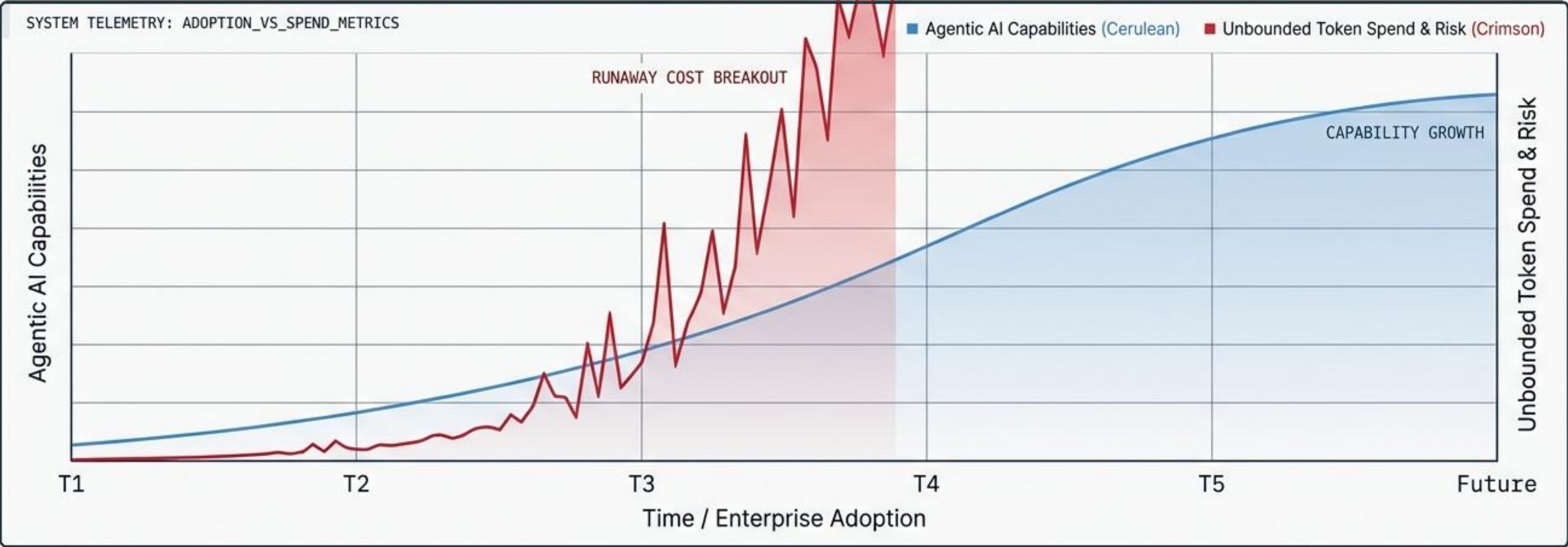


THE AI TOKEN CONTROL PLANE

Governance, Optimization, and FinOps for Enterprise Agentic AI



TOKEN CONSUMPTION IS THE NEW COMPUTE COST



WIDGET_ID: PROBLEM_STATEMENT_001

THE PROBLEM

Uncontrolled AI adoption leads to excessive use of frontier models, duplicate requests across teams, and runaway agentic workflows.

STATUS: CRITICAL | RISK: HIGH | COMPONENT: ORG_BEHAVIOR

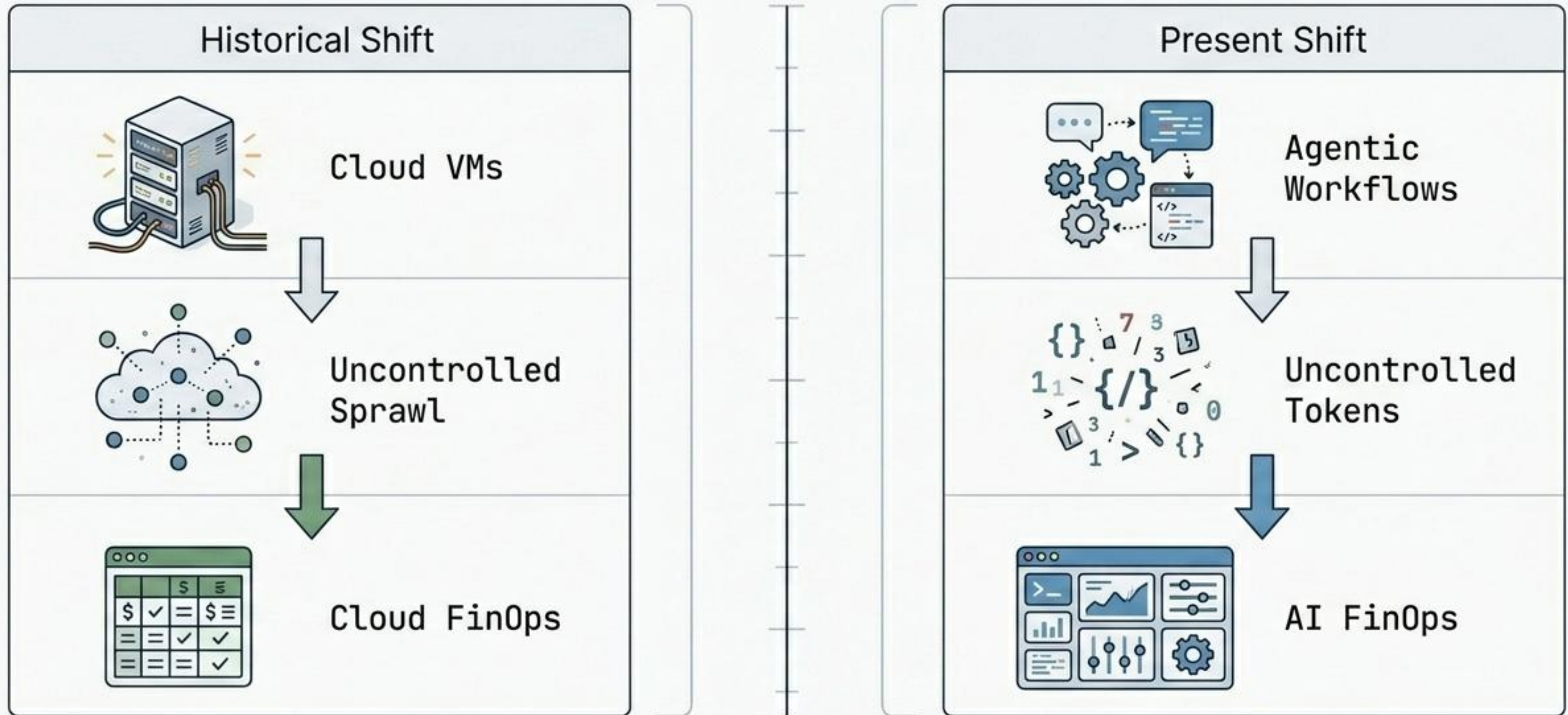
WIDGET_ID: REALITY_CHECK_002

THE REALITY

Organizations face rapidly growing AI operating costs disconnected from business outcomes, with zero cost attribution or governance.

STATUS: ALERT | FINANCIAL_IMPACT: SEVERE | COMPONENT: GOVERNANCE_GAP

Agentic AI requires AI FinOps.



The Mandate: You cannot optimize what you do not route and meter. Make every LLM call **metered**, **routed**, **cached**, compressed, **governed**, and **priced by outcome**.

A single governed gateway sits in front of every LLM call.



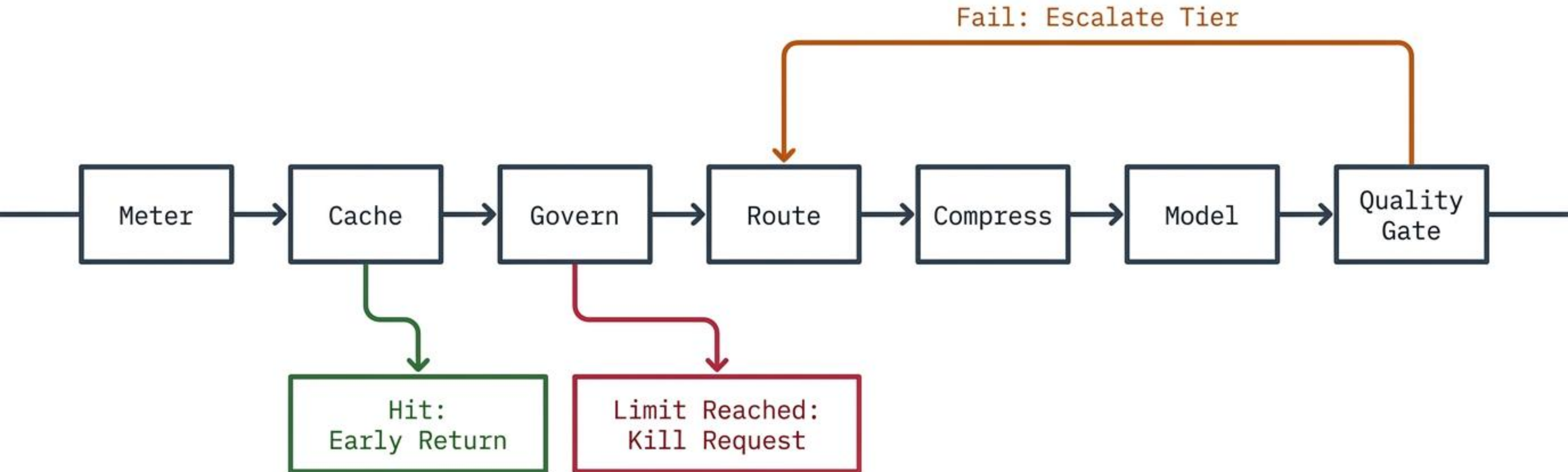
The Mechanism	The Strategy
Acts as an OpenAI-compatible proxy. It is the one place that can see, cut, cap, and price token spend across the organization.	Centralized operation ensures no naked API calls bypass enterprise policy.

A control plane is not just a compressor.

Capability	Naked API Calls	Payload Compressors	AI Token Control Plane
Shrinks Payloads			
Selects Optimal Model			
Governs Budgets			
Stops Runaway Agents			
Attributes Cost by Outcome			

The Moat: Compression is just one pluggable stage. The defensible enterprise layer is routing, governance, and outcome pricing.

The hot path isolates and eliminates waste at every stage.



Note: Everything to the right of the gateway (stores, telemetry, FinOps) happens asynchronously to preserve low latency on the hot path.

Four levers dynamically reduce tokens, cost, and risk.

Semantic Cache

Cuts Tokens via reuse.

Skips external model calls entirely based on embedding match.

Intelligent Router

Cuts Cost via cheaper tiers.

Selects the cheapest model capable of clearing the quality bar.

Compression Adapter

Cuts Tokens via smaller payloads.

Applies content-aware context shrink to the prompt footprint.

Budget Governor

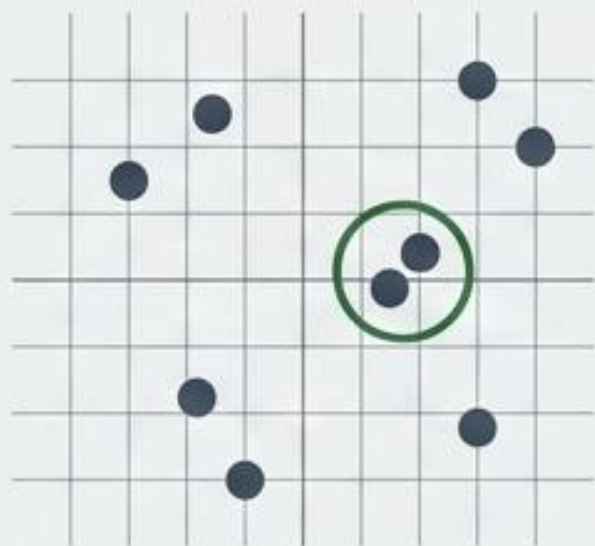
Cuts Risk via blast radius caps.

Enforces per-run limits and activates runaway loop kill switches.

Three levers structurally reduce the bill. One lever physically caps enterprise risk.

Optimization Deep Dive: Semantic Reuse and Cold-Start Routing

Semantic Cache



```
store: pgvector / FAISS  
rule: cosine_similarity >= 0.95  
action: return cached_answer
```

Dynamically returns prior answers on exact or semantic match to bypass compute.

Router

Task: codegen
capability_required: 0.85

Small Tier:
0.70 capability

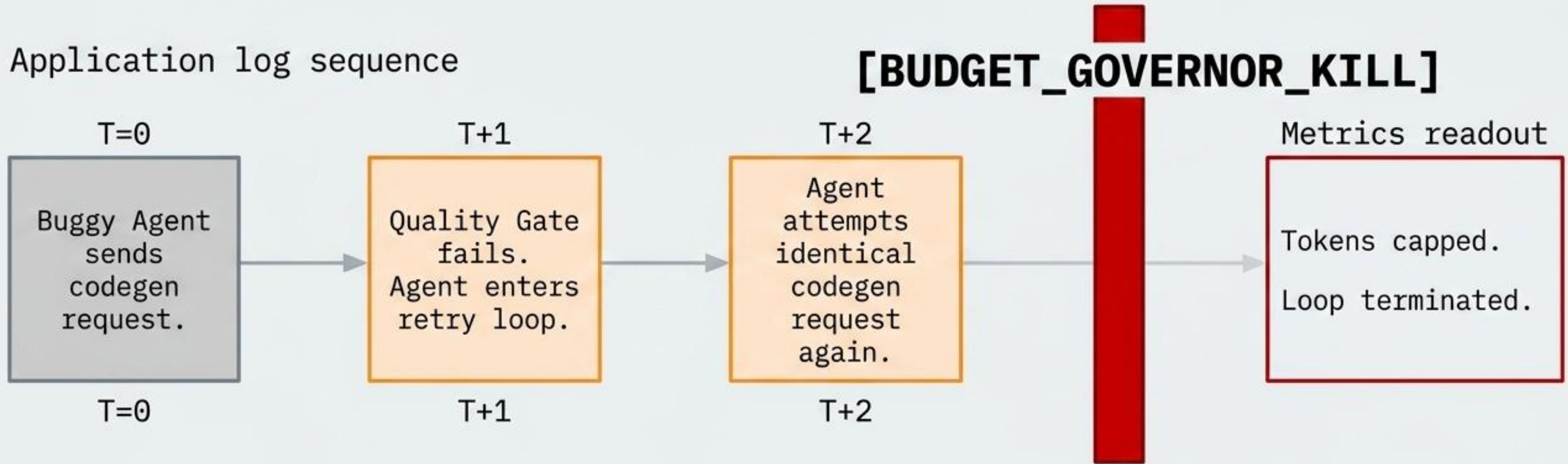
Mid Tier:
0.88 capability

Frontier Tier:
0.97 capability

Automatically routes to the cheapest capable tier, saving Frontier models for only the hardest tasks.

Governance Deep Dive: The Runaway Kill Switch

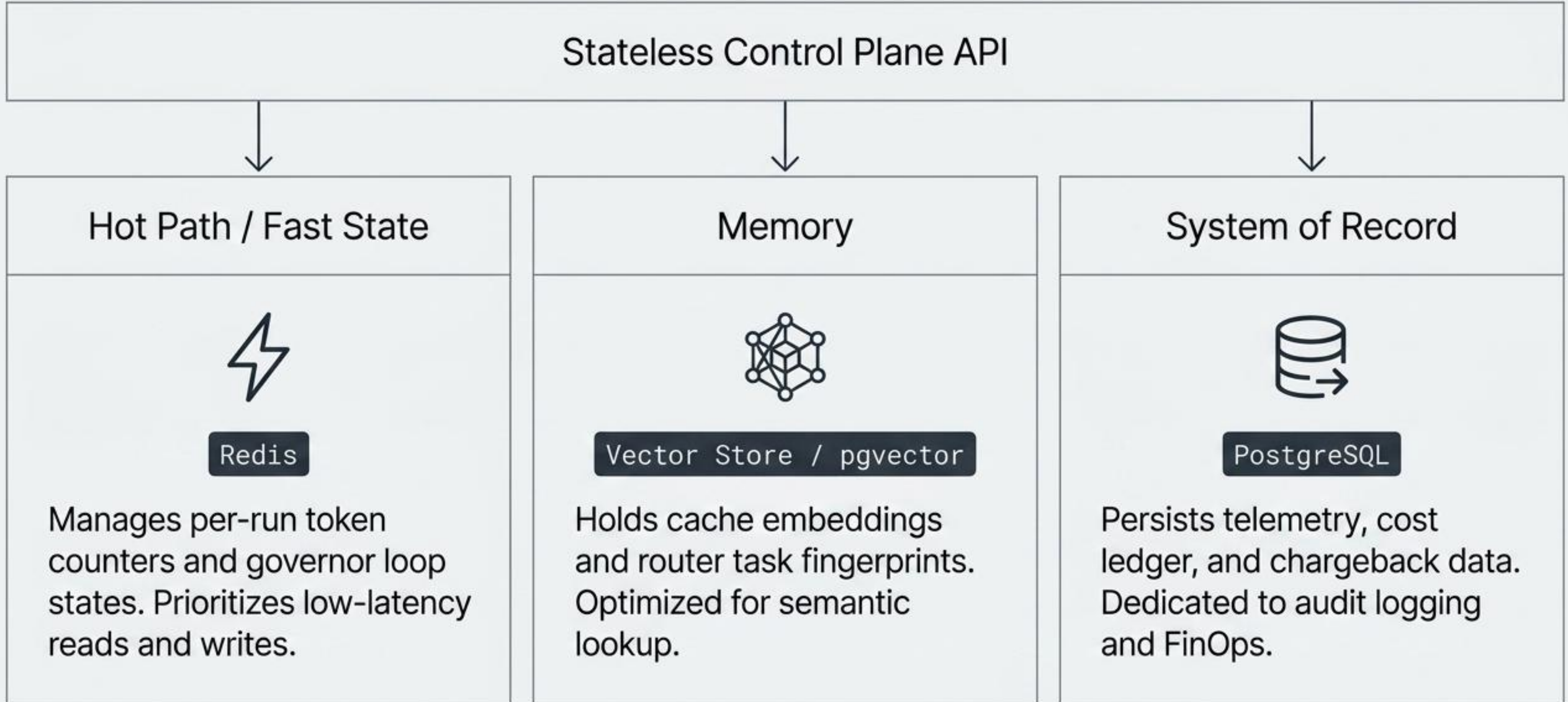
Application log sequence



Scenario: A buggy coding agent stuck in a retry loop.

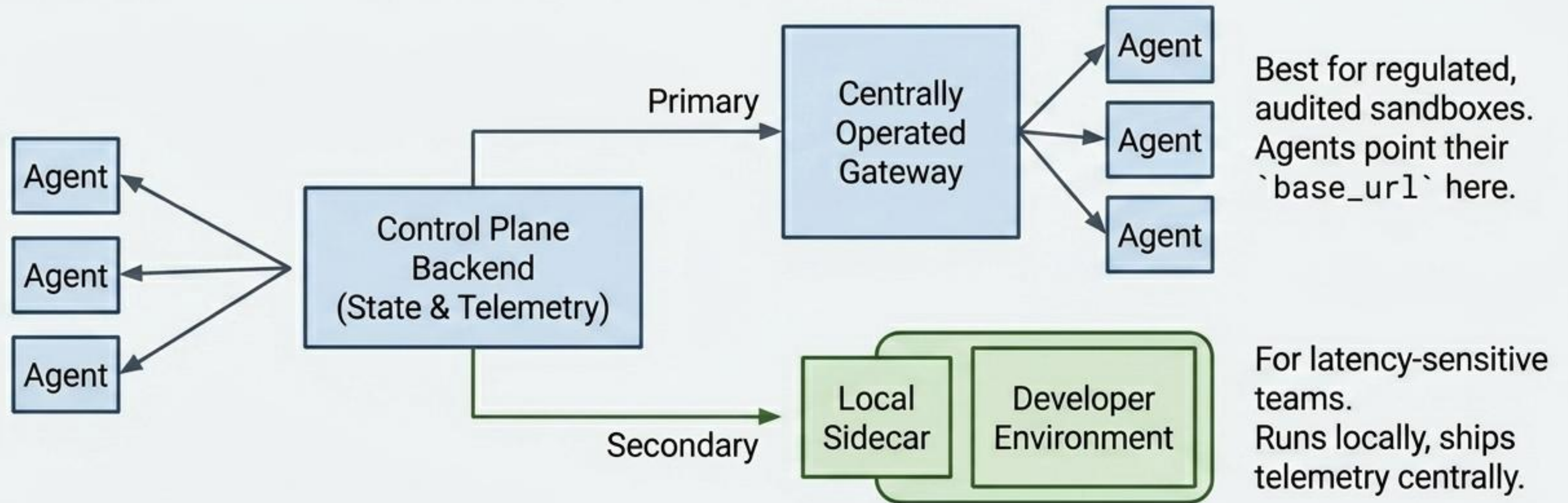
Outcome: The governor limits blast radius, prevents infinite token spend, and automatically protects shared enterprise infrastructure.

A stateless gateway powered by externalized memory and ledgers.



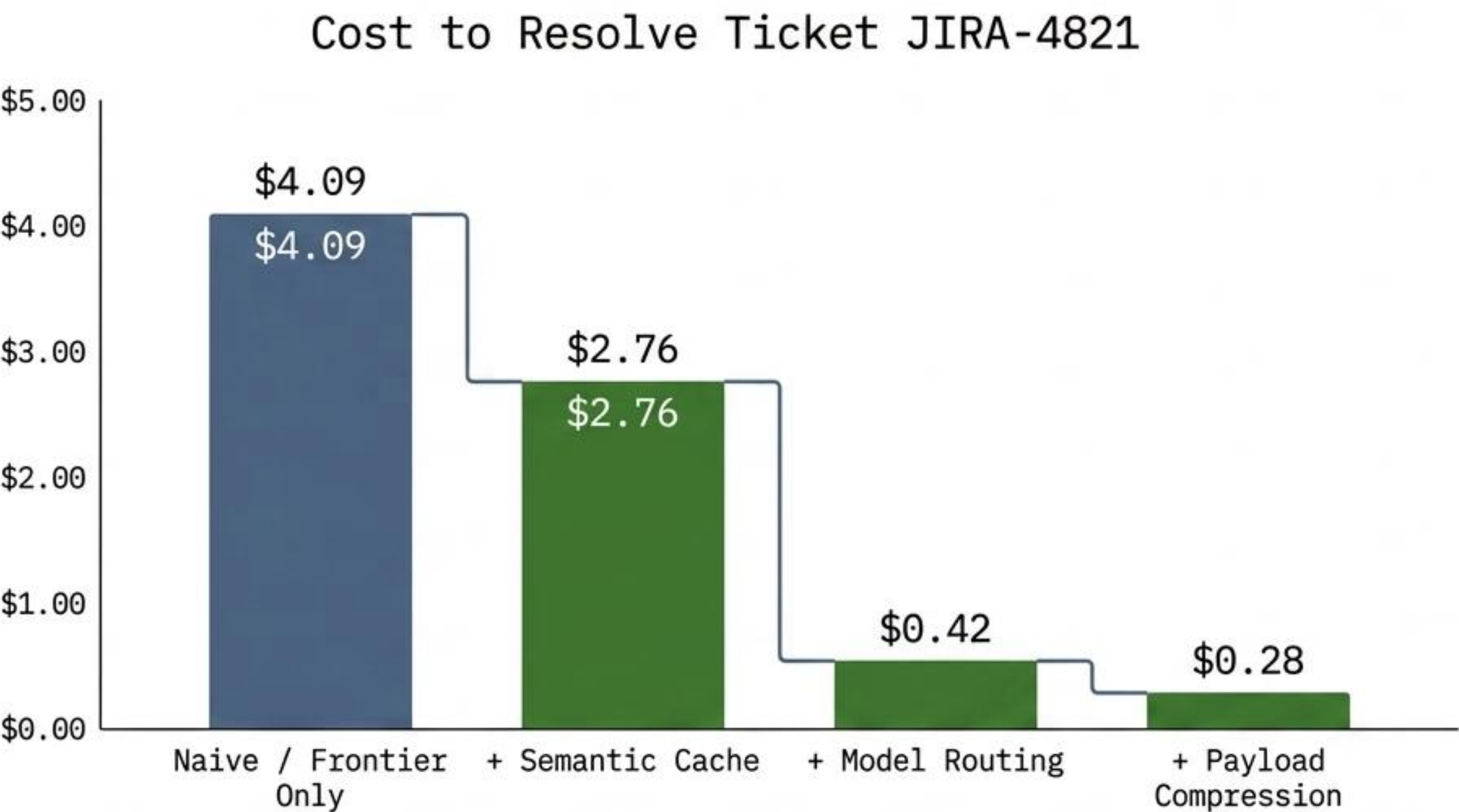
Flexible deployment topologies within the enterprise trust boundary.

Enterprise Trust Boundary: No proprietary payloads leave the network.



- **Data Residency:** Compression and semantic caching execute securely inside your network.
- **RBAC & Quotas:** Per-team budgets are enforced centrally, regardless of the invocation origin.

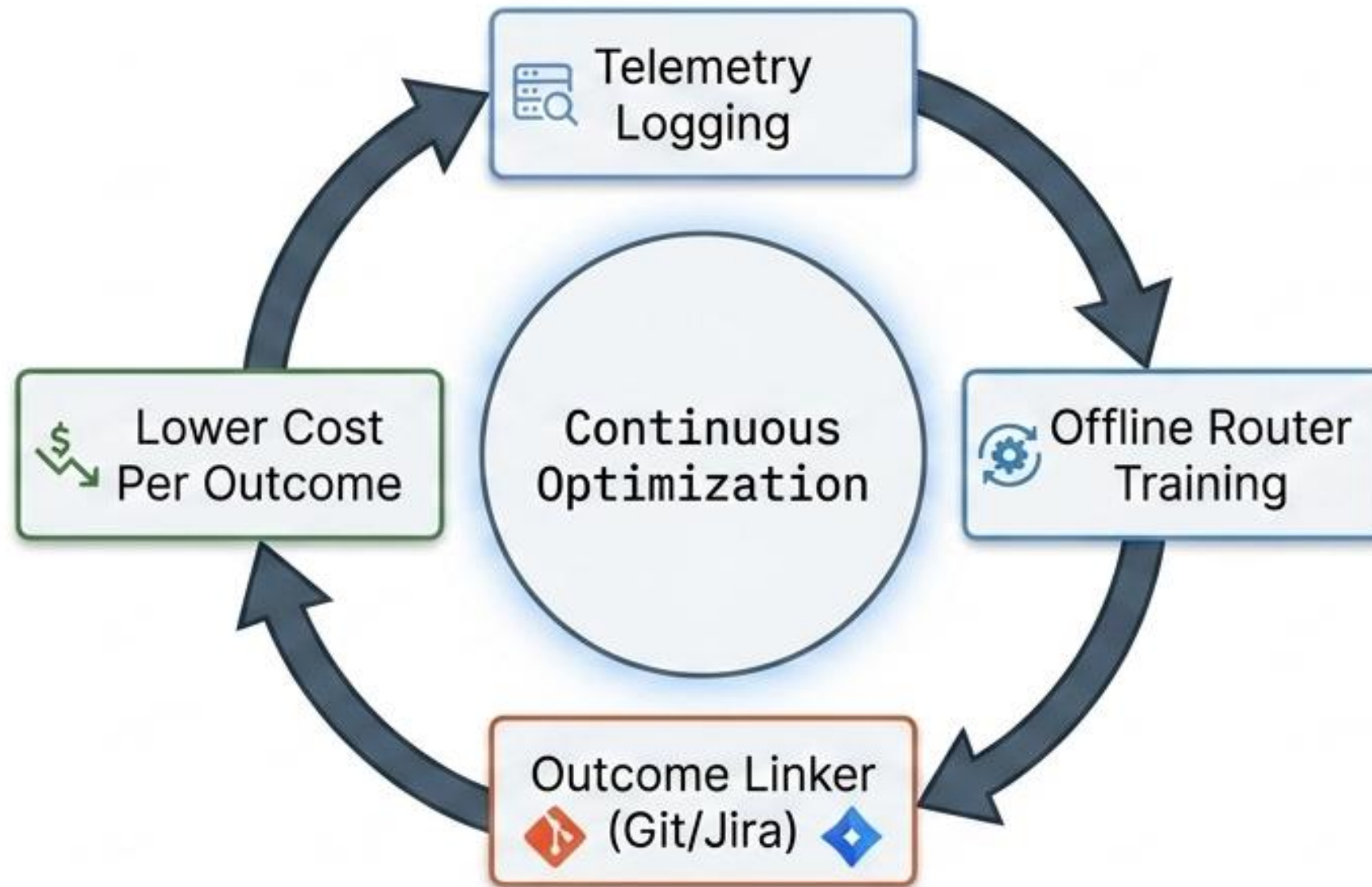
Verified Impact: 93% cost reduction, quality maintained.



Total Tokens
273,000 to 114,430 (-58%)
Quality Score
0.93 to 0.89 (Maintained)
QA Cost Per Outcome
\$4.38 to \$0.31 (-93%)

Each optimization layer compounds. Quality is preserved while unit economics are fundamentally transformed.

The foundation for enterprise-scale AI.



The Future State:

The enterprise moat is outcome pricing tuned on proprietary telemetry.

The Mandate:

Make every LLM call metered, routed, cached, compressed, governed, and priced by outcome.
Scale AI adoption without costs scaling with it.